

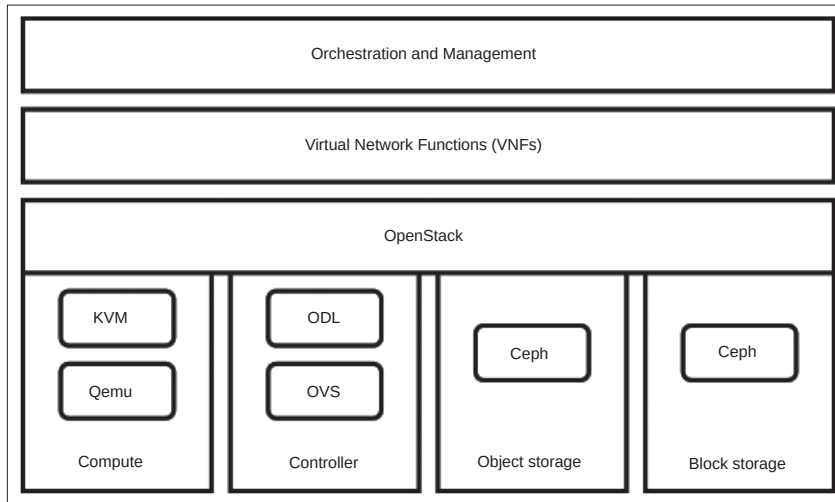


Figure 2: NFV infrastructure management with OpenStack

little too old. Operating system virtualization is a budding virtualization technology and is much more lightweight than that of VMs. Then there are containers that enable microservices, where many physical network functions migrate to virtual network functions (VNFs). As of today, these VNFs are transforming and migrating to cloud native networking functions (CNFs), which are implementing network functionalities that coexist through one or more microservices. CNFs are boosting the telecom industry with cutting-edge network architecture integrated with DevOps processes, aiming for greater service flexibility and a faster delivery mechanism.

Where is Kubernetes in all of this?

We know Kubernetes as an orchestration tool, which automates the entire deployment and scaling procedure of container-based applications. But in relation to NFV, it also covers virtual infrastructure management. In this role, it schedules container-based workloads and jobs while managing the networking between them.

Kubernetes can manage virtual network functions by monitoring the health of each container and restarting these when required. Operators can also scale the containers up or down based on the incoming traffic.

The grand scheme of NFV is compelling in cloud native computing.

Today, Kubernetes is bringing quite a change in NFV infrastructure, but the evolution from VMs to container-based environments is still a challenge. While the operator might deny it today, surely this is the need of the hour going by the automation, scaling, and monitoring functionalities Kubernetes brings to the table. Without any doubt, this is the path to the future of NFV infrastructure management.

NFV infrastructure is difficult to manage in an environment where developers and operators clash. This can lead to slow movement of NFV applications on the infrastructure. Apart from this, the setting up of such infrastructure is a challenge. DevOps methodologies are a buzz in the market and are known for their automated tools chain. They also help connect developers with the production environment of the Kubernetes cluster and OpenStack cloud services. **END** 🐙

References

- Margaret Chiosi, Don Clarke, Peter Willis, Andy Reid, James Feger, Michael Bugenhagen, Waqar Khan, Michael Fargano, Chunfeng Cui, Hui Deng, et al; Network Functions virtualisation: An introduction, benefits, enablers, challenges and call for action; In SDN and OpenFlow World Congress, volume 48, pages 1–16, 2012
- Ken Gray and Thomas D. Nadeau; Network Function Virtualization; Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 1st edition, 2016; ISBN:0-12-802119-5
- Hassan Hawilo, Abdallah Shami, Maysam Mirahmadi, and Rasool Asal; NFV: state of the art, challenges, and implementation in next generation mobile networks (vEPC); IEEE Network, 28(6):18–26, 2014; DOI:10.1109/MNET.2014.6963800.
- OpenStack for Telecom & NFV; <https://www.openstack.org/use-cases/telecoms-and-nfv/>; Last accessed: March 1, 2022
- Russell Bryant, Kathy Cacciatore, Stephen Gordon, Eric Ji, Armando Migliaccio, Iben Rodriguez, et al; Accelerating NFV Delivery with OpenStack; <https://object-storage-ca-ymq-1.vexxhost.net/swift/v1/6e4619c416ff4bd19e1c087f27a43eea/www-assets-prod/marketing/OpenStack-NFV-A4.pdf>; Last accessed: March 1, 2022
- Fabio V; How Kubernetes is Helping NFV Towards Cloud Native; <https://gonorthforge.com/how-kubernetes-is-helping-nfv-towards-cloud-native/>; Last accessed: March 15, 2022

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